THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 5 & 6: **Life Processes, Control and Coordination** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Glomerulus is a cluster of: 1

(a) lymph vessels **(b)** nerve cells
(c) blood capillaries **(d)** muscle cells

2. Which enzyme converts fats into fatty acids and glycerol? 1

(a) lipase **(b)** trypsin **(c)** pepsin **(d)** amylase

3. In single-celled organisms, gas exchange occurs by: 1

(a) tracheae **(b)** lungs **(c)** diffusion **(d)** gills

4. The pacemaker of the heart is located in the: 1

(a) left atrium **(b)** left ventricle **(c)** right atrium **(d)** aorta

5. The opening and closing of stomata is controlled by: 1

(a) phloem **(b)** guard cells **(c)** epidermal cells **(d)** xylem

6. The gland located in the neck is: 1

(a) pituitary **(b)** pancreas **(c)** adrenal **(d)** thyroid

7. Oestrogen is secreted by: 1

(a) pancreas **(b)** thyroid **(c)** testes **(d)** ovaries

8. Which plant hormone is mostly responsible for closing of stomata in drought? 1

(a) gibberellin **(b)** auxin **(c)** abscisic acid **(d)** cytokinin

9. The part of the neuron where information is acquired is: 1

(a) nerve ending **(b)** axon **(c)** myelin **(d)** dendrite

10. Picking up a pencil is controlled by the: 1

(a) cerebellum **(b)** medulla **(c)** spinal cord **(d)** hypothalamus

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Name the end products of digestion of carbohydrates, proteins and fats (any two). 2

12. Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds? 2

13. Name the hormone released by the thyroid and its deficiency disease. 2

14. What are effectors? Give one example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What is the importance of transpiration in plants? 3

16. What are the effects of hormones secreted at puberty? Name the hormones. 3

17. Name the parts of the human heart which 3
(i) receive blood from the body
(ii) pump blood to the lungs
(iii) receive oxygenated blood.

OR

Why is diffusion insufficient to meet the oxygen requirement of multicellular organisms? 3

18. State the functions of the pituitary gland. Why is it called the master gland? 3

OR

Explain the role of the hypothalamus in hormone secretion. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is photosynthesis? Write its equation. 5
(ii) Explain the role of stomata.
(iii) What happens to excess glucose?

OR

(i) Describe the mechanism of breathing in humans. 5
(ii) Why is the residual volume of air in the lungs important?

20. (i) Explain chemotropism with an example. 5
(ii) Explain thigmotropism with an example.
(iii) Differentiate tropic and nastic movements.

OR

(i) What is the role of adrenaline? 5
(ii) Where is it secreted?
(iii) Explain how the body responds in an emergency.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND